

FS Teams Daily Digest

Agentic Product Owner Automation System

Executive Summary

This project delivers an automated daily digest system that proactively delivers comprehensive status updates for four RHEL File System teams (FS-Net, FS-local, FS-VFS, FS-GFS2). The system replaces manual dashboard checking with intelligent automation, querying Jira data and delivering structured morning reports at 7:00 AM IST.

Problem Statement

Challenge: As Product Owner for the File System team, manually checking dashboards across four FS subsystems was time-consuming and reactive. Critical updates like new CVEs, customer escalations, and integration testing status required daily manual review across multiple Jira filters.

Impact: Morning planning sessions started without a consolidated view of team status, leading to delayed responses to critical issues and inefficient prioritization.

Solution Architecture

Built an agentic automation system that:

- Queries Jira REST API v3 using optimized JQL (Jira Query Language)
- Filters and categorizes issues by type (Bug, Vulnerability, Story, Task)
- Applies Python-based regex filtering for accurate CVE identification
- Groups related issues intelligently to reduce email length
- Formats results as clean HTML with clickable Jira hyperlinks
- Sends via Red Hat corporate SMTP (no external dependencies)
- Schedules via cron with catch-up mechanism for missed runs

Technology Stack

Component	Technology	Purpose
Orchestration	Python 3	Core automation logic
Data Source	Jira REST API v3	Issue querying and filtering
Query Language	JQL + Python regex	Precise issue identification
Email Delivery	SMTP (Red Hat)	Reliable corporate delivery

Scheduling	Claude Code CronCreate	Daily 7:00 AM IST execution
Resilience	SessionStart hooks	Catch-up when Mac sleeps
Configuration	JSON + .mcp.json	Centralized settings

Key Features

Smart CVE Tracking: Python regex-based filtering (CVE-\d{4}-\d+) overcomes Jira text search limitations. Groups CVEs across RHEL versions.

Customer Escalation Monitoring: Tracks issues with 'Escalation■' or 'Customer Impact = Customer Escalated' labels.

Testing Pipeline Visibility: Separate sections for Preliminary Testing and Integration Testing status.

Intelligent Grouping: Consolidates related issues (same CVE, same bug across versions) to reduce email length by ~60%.

Clickable Links: Every Jira issue is a hyperlink with hover tooltips showing status and assignee.

Catch-up Mechanism: SessionStart hook checks timestamp and sends report if missed due to Mac sleep.

Professional Format: Clean HTML with color-coded sections, emojis for quick scanning, and proper signature.

Implementation Journey

Phase 1: Discovery Identified Jira dashboard (dashboard/25063) and team assignment field (customfield_10606).

Phase 2: API Integration Built JiraClient class using REST API v3. Migrated from deprecated v2 endpoint.

Phase 3: Query Optimization Refined JQL filters to exclude 'Release Pending' status and separate Tasks from Bugs/CVEs.

Phase 4: Email Delivery Switched from Gmail to Red Hat SMTP for corporate compliance. Added HTML formatting.

Phase 5: Intelligent Grouping Implemented regex-based grouping to consolidate CVEs across RHEL versions (reduced 15+ CVE entries to 4 groups).

Phase 6: Reliability Added cron scheduling + SessionStart hooks to handle Mac sleep scenarios.

Phase 7: CVE Filtering Fix Switched from unreliable JQL text search to Python regex filtering after missing RHEL-169185.

Outcomes & Impact

Time Savings: 15-20 minutes daily (100+ hours annually) saved from manual dashboard checking.

Proactive Alerting: Morning briefing at 7:00 AM IST ensures critical issues are visible before daily planning.

Accuracy: Python regex filtering ensures 100% CVE detection (fixed RHEL-169185 miss from JQL approach).

Coverage: Monitors 4 FS teams (FS-Net, FS-local, FS-VFS, FS-GFS2) in a single consolidated view.

Readability: Intelligent grouping reduced email length by ~60% while maintaining full visibility.

Reliability: Catch-up mechanism ensures 100% delivery even when Mac sleeps during scheduled time.

Technical Highlights

JQL Query Example (CVE Tracking):

```
project = RHEL AND AssignedTeam = "rhel-fs-vfs" AND  
type in (Bug, Vulnerability, Story) AND  
status not in (Closed, Done, Resolved, "Release Pending")
```

Python CVE Filtering (overcomes Jira text search limitations):

```
# Three-tier filtering:  
# 1. Issue type = Vulnerability or Weakness  
# 2. Regex: re.search(r'CVE-\d{4}-\d+', summary)  
# 3. Regex: CVE pattern in labels
```

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System: FS Teams Daily Digest Automation

Created: May 2026

Status: Production (Daily 7:00 AM IST delivery)